NIHARIKA JAVVAJI

SOFTWARE ENGINEER

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Accomplished **Software Engineer with over 3.8 years of experience** designing and deploying scalable, Machine Learning, Deep Learning, and Agentic AI solutions. Proficient in **Python, SQL, Pandas, NumPy, and Scikit-learn**, with a proven ability to translate complex datasets into high-value business strategies. Dedicated to **continuously acquiring knowledge in emerging AI paradigms** to foster cross-functional collaboration, solve critical business problems, and drive technical innovation.

PROFESSIONAL EXPERIENCE

➤ CAPGEMINI TECHNOLOGIES SERVICE Pvt. Ltd., Hyderabad **Dec 2022 - Present**

CLIENT: RBC , Credit Risk Scoring

- Designed **credit risk scoring models** using **bank statement, credit bureau, and Days Past Due (DPD)** data to accurately assess customer creditworthiness.
- Optimized a **high-throughput scoring pipeline** processing **300–400 bank statements** per minute, reducing end-to-end runtime by 10%.
- Built and enhanced **Savings Account (SA)** and **Current Account (CA)** scoring models, improving **ROC-AUC by up to 14%** and **Gini coefficient by up to 17%** through feature engineering and model optimization.
- Developed a **New-to-Credit (NTC)** machine learning model using alternative financial and transaction data, achieving a **ROC-AUC of 0.82** and **Gini coefficient of 0.64**, while supporting model validation and **model deployment,** and **optimize data pipelines**

➤ LLM-Powered Enterprise Document Intelligence System

- Contributed to an **LLM-powered Enterprise Knowledge Assistant** using **RAG, LangChain, Gemini, ChromaDB,** and **Hugging Face embeddings** for intelligent enterprise document search and Q&A.
- Collaborated with senior engineers on the **AI architecture** and **end-to-end RAG pipeline**, including **document ingestion, chunking, embeddings, vector storage, retrieval, and response generation**.
- Implemented **semantic search, contextual retrieval, and reranking** to improve the relevance of information provided to the LLM.
- Added **source citations, hallucination prevention, prompt engineering, and response validation** to improve answer reliability, traceability, and factual grounding.

PROJECTS

➤ LLM-Powered Retail Sales Analytics Chatbot

Developed an LLM-powered retail sales analytics chatbot using **Python**, **SQLite**, **Gradio**, and **Groq LLM enabling retail managers and business analysts to monitor sales performance**, identify top-selling products, analyze customer purchasing trends.

➤ AI-Powered Regulatory & Tax Compliance Assistant using LangGraph, RAG & Agentic AI

Developed an AI-powered regulatory compliance assistant using **LangGraph and RAG** to answer queries from **government regulations and policy documents**.

Implemented **FAISS, Hugging Face BGE embeddings, reranking, source citations, hallucination prevention**, and **intelligent tool routing** for accurate, explainable responses.

➤ Hourly Traffic Volume Forecasting System | Uber Mentor-Led Internship

Developed an hourly time-series traffic forecasting model during an Uber mentor-led internship program. Engineered **temporal features** and **tuned XGBoost, Scikit-learn** to predict junction volume from **historical traffic logs**. Achieved a **15–20% reduction in forecasting error (RMSE/MAE)** and can be integrated into **smart city platforms to optimize dynamic traffic signal timing** and power real-time navigation rerouting.

➤ BERT-Based Abstractive Text Summarization

Developed an **NLP application** to generate concise summaries using a **BERT-based transformer** model. Implemented text **preprocessing, tokenization,** and **model fine-tuning**, evaluating performance with **ROUGE** metrics to produce coherent and context-aware summaries. This can **automatically condense lengthy reports**, articles, business documents into **concise summary.**

➤ Dog Emotion Recognition using MobileNet and Transfer Learning

Developed a **canine emotion recognition system using CNN, MobileNet, and transfer learning** emotions from facial images. Implemented image preprocessing, data augmentation, **CNN fine-tuning**, and performance evaluation to improve classification accuracy designed for integration with **robotic and smart pet-care systems to enable real-time emotion monitoring** and responsive pet interaction

SKILLS

- **Programming & Data:** Python, Java, C, SQL (MySQL), NumPy, Pandas
- **Machine Learning & Deep Learning**: Machine Learning, Deep Learning, Scikit-learn, TensorFlow, PyTorch, NLP, Computer Vision, MLflow
- **Generative AI & LLMs:** LangChain, LangGraph, Retrieval-Augmented Generation (RAG), Model Context Protocol (MCP), CrewAI, AutoGen, Hugging Face Transformers, Ollama, FAISS
- **Backend & Cloud:** FastAPI, Flask, Docker, REST APIs, Amazon Web Services (AWS) , Git, GitHub

EDUCATION

Bachelor of Technology (B. Tech, CSE) — **2018 - 2022**
St. Peter's Engineering College

Intermediate — **2016 - 2018**
Sri Chaitanya Jr College

Secondary School — **2016**
K.V.M Talent High School

CERTIFICATIONS

- Deep Learning Specialization – Coursera (DeepLearning.AI)
- Machine Learning Specialization – Coursera (DeepLearning.AI)
- Professional Certificate Program in AI & Data Science – upGrad (in association with PwC Academy)
- Uber Mentor-Led Internship Program